

SIMATS ENGINEERING

ASSESSMENT TOOL 1- Enumeration Analysis

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO3: *Analyse network services using port scanning, ping sweeps, and enumeration techniques across different operating systems to identify vulnerabilities. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:100

1. A cyber security consulting company has been contracted to perform an initial security assessment of the publicly accessible server **scanme.nmap.org**. Before conducting any vulnerability assessment, the client requires a detailed enumeration report. Using **Nmap**, perform host discovery, TCP port scanning, service version detection, operating system fingerprinting, and default script scanning on **scanme.nmap.org**. Analyze all identified ports and services, determine their business purpose, and evaluate whether they are necessary for public access. Discuss how a threat actor could utilize the information gathered during enumeration to launch further attacks such as service exploitation, brute-force attacks, or vulnerability discovery. Prepare a technical report summarizing your findings, attack surface analysis, risk assessment, and recommendations for securing the server.
2. The security team has deployed a deliberately vulnerable Linux server, **192.168.56.101 (Metasploitable 2)**, in an isolated laboratory environment to train penetration testers. As part of the pre-engagement reconnaissance phase, use **Nmap** to perform comprehensive enumeration of **192.168.56.101**. Identify all open ports, running services, service versions, and operating system information. Analyze the purpose of each service and determine which services could provide potential entry points for an attacker. Categorize the discovered services into High, Medium, and Low risk levels based on their exposure and functionality. Explain how the enumeration results could guide a penetration tester in selecting potential attack vectors and exploitation strategies.
3. A financial organization wants to assess the security posture of a Windows workstation located at **192.168.56.102** before connecting it to the production network. Using **Nmap**, enumerate **192.168.56.102** to identify active ports, network services, file-sharing protocols, remote management services, and operating system details. Analyze whether the discovered services comply with the principle of least privilege and identify any services that may unnecessarily increase the attack surface. Discuss how an attacker could leverage the identified services for reconnaissance, lateral movement, or unauthorized access. Based on your findings, propose a set of security hardening measures to improve the system's resilience against cyberattacks.

4. An organization hosts a web application on **192.168.56.103 (OWASP Juice Shop Server)** and plans to conduct a full penetration test. Before the testing phase begins, the security team must understand the server's network exposure. Using **Nmap**, perform a detailed enumeration of **192.168.56.103**, including host discovery, port scanning, service version detection, operating system fingerprinting, and web service identification. Analyze the discovered services and determine how they support the hosted web application. Evaluate the risks associated with exposed web services, administrative interfaces, and supporting services. Explain how the information gathered during enumeration could be used to identify potential vulnerabilities in the web application environment and recommend measures to minimize exposure
5. The network gateway device at **192.168.56.1** serves as the primary access point between the internal laboratory network and external networks. The network administrator suspects that unnecessary management services may be exposed. Using **Nmap**, conduct a comprehensive enumeration of **192.168.56.1**, including port scanning, service detection, operating system fingerprinting, and script-based enumeration. Identify all exposed management interfaces, network services, and accessible resources. Analyze the security implications of exposing services such as SSH, Telnet, SNMP, HTTP, or HTTPS management interfaces. Discuss how attackers could exploit misconfigured management services to gain unauthorized access to the network infrastructure. Based on your analysis, recommend security controls and best practices to secure the gateway device and reduce the risk of compromise.

Code of Conduct:

I Mohan Kumar J (192424060) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Mohan Kumar J

Mohan Kumar J · PROJECT-001
WM-PDF-2026-54365CC3

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Mark
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts	
Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model	
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools	
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results	
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation	

1) Nmap Enumeration Assessment Review

Performed **Nmap enumeration** on the public server **scanme.nmap.org** to identify active hosts, open TCP ports, running services, service versions, operating system details, and default NSE script results.

The scan identified publicly accessible services such as **HTTP, SSH, and other network services**, which were analyzed to determine their business purpose and whether they were necessary for public access.

The discovered services helped evaluate the server's **attack surface**, highlighting services that could become potential entry points if left outdated or misconfigured.

An attacker could use the enumeration results to perform **service exploitation, brute-force attacks, or vulnerability discovery** by targeting exposed services and comparing detected versions with known vulnerabilities.

To improve security, it is recommended to **disable unnecessary services, keep software updated, enforce strong authentication, restrict remote access, use firewalls, and perform regular vulnerability assessments and security monitoring**

```
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:09 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.824x latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c::9fff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

```
[kali@kali]-[~]
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:10 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.824x latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c::9fff:fe18:b02f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
513/tcp   closed mysql
80/tcp    open  http
Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
144/tcp   closed pop3
3128/tcp  closed squid-http
3189/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8085/tcp  closed unknown
8089/tcp  closed unknown
Device type: bridge/vpn
Running (OST GUESTS): Oracle VirtualBox (94%), SLIRP (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox;cpe:/a:mam-gasparov:silirp;cpe:/q:qemu
Aggressive OS guesses: Oracle VirtualBox NAT Bridge (94%), AT&T iWDM voice gateway (92%), QEMU user mode ne
twork gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

```
TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 0.44 ms scanme.nmap.org (45.33.32.156)
```

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

```
Nmap done: 1 IP address (1 host up) scanned in 100.67 seconds
```

```
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:09 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.824x latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c::9fff:fe18:b02f
Nmap done: 1 IP address (1 host up) scanned in 0.64 seconds
```

```
[kali@kali]-[~]
$ sudo nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-06 05:10 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.824x latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c::9fff:fe18:b02f
Not shown: 987 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
513/tcp   closed mysql
80/tcp    open  http
Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
143/tcp   closed imap
144/tcp   closed pop3
3128/tcp  closed squid-http
3189/tcp  closed ms-wbt-server
5060/tcp  open  sip
5061/tcp  open  sip-tls
8080/tcp  closed http-proxy
8085/tcp  closed unknown
8089/tcp  closed unknown
Device type: bridge/vpn
Running (OST GUESTS): Oracle VirtualBox (94%), SLIRP (94%), AT&T embedded (92%), QEMU (90%)
OS CPE: cpe:/o:oracle:virtualbox;cpe:/a:mam-gasparov:silirp;cpe:/q:qemu
Aggressive OS guesses: Oracle VirtualBox NAT Bridge (94%), AT&T iWDM voice gateway (92%), QEMU user mode ne
twork gateway (90%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

OS and Service detection performed. Please report any incorrect results at <https://nmap.org/submit/>.

```
Nmap done: 1 IP address (1 host up) scanned in 100.67 seconds
```

2) Performed Nmap enumeration on the Metasploitable 2 server (192.168.56.101) to identify open ports, running services, service versions, and operating system details.

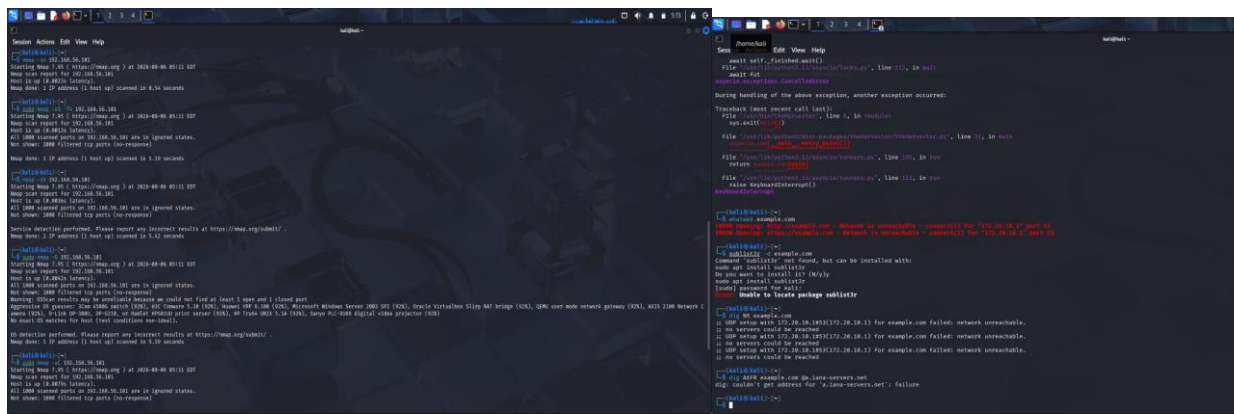
The scan discovered multiple services such as FTP, SSH, Telnet, HTTP, SMTP, MySQL, and Samba, indicating several publicly accessible entry points.

Services were categorized based on risk:

- High Risk: FTP, Telnet, Samba (known vulnerabilities and insecure protocols)
- Medium Risk: SSH, HTTP (secure when properly configured but still require monitoring)
- Low Risk: Other informational or less exposed services.

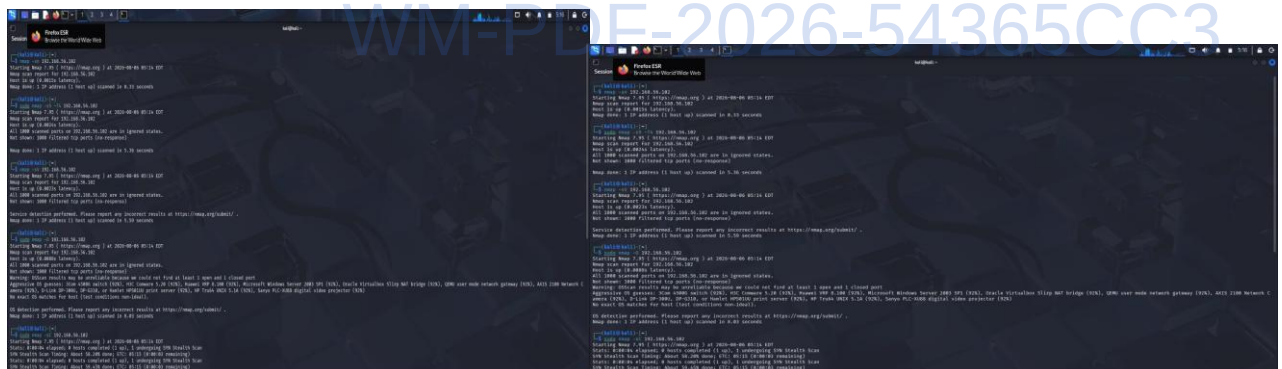
The enumeration results help penetration testers identify potential attack vectors, prioritize vulnerable services, and select suitable exploitation strategies.

Regular patching, disabling unnecessary services, enforcing strong authentication, and continuous monitoring are recommended to reduce the server's attack surface and improve overall security.



3) Performed **Nmap enumeration** on the Windows workstation (**192.168.56.102**) to identify open ports, network services, file-sharing protocols, remote management services, and operating system details.

- The scan identified services such as **SMB (file sharing), RDP (Remote Desktop), RPC, NetBIOS, and other Windows network services**, which are commonly used for system administration and resource sharing.
- Services were evaluated based on the **principle of least privilege**. Unnecessary or publicly exposed services increase the system's attack surface and should be disabled or restricted.
- An attacker could use the discovered services for **reconnaissance, lateral movement**, password attacks, or unauthorized remote access if weak credentials or vulnerabilities exist.
- To improve security, it is recommended to **disable unused services, restrict SMB and RDP access, enable the Windows Firewall, apply security patches, enforce strong passwords with Multi-Factor Authentication (MFA), and regularly monitor system logs** to reduce the risk of cyberattacks.

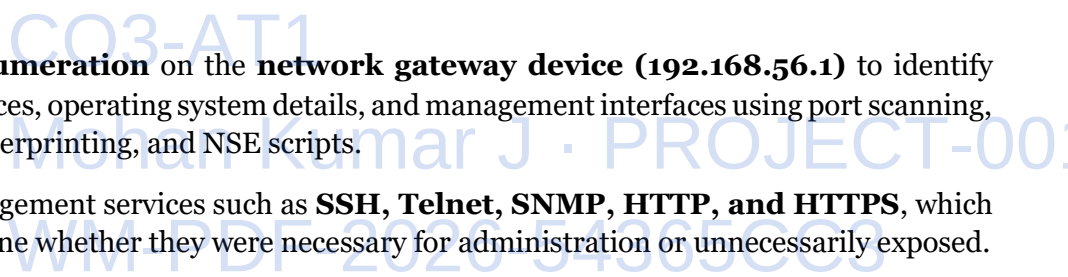


4) Performed **Nmap enumeration** on the **OWASP Juice Shop Server (192.168.56.103)** to identify active hosts, open ports, running services, service versions, operating system details, and web services.

The scan identified **HTTP/HTTPS web services** along with supporting network services required for hosting and managing the web application. These services were analyzed to understand their role in supporting the application.

Exposed web services and administrative interfaces may increase the **attack surface** if they are misconfigured, outdated, or accessible without proper security controls.

To improve security, it is recommended to **disable unnecessary services, restrict access to administrative interfaces, keep web server software updated, enable HTTPS, configure firewalls, and perform regular vulnerability assessments and security monitoring.**



The scan identified management services such as **SSH, Telnet, SNMP, HTTP, and HTTPS**, which were analyzed to determine whether they were necessary for administration or unnecessarily exposed.

An attacker could exploit these services to gain **unauthorized access, perform reconnaissance, brute-force login credentials, gather network information, or compromise the gateway**, potentially affecting the entire network.

[illegible]